

Lecture 10b

- market demand
- intro to equilibrium

Market Demand

- recall: by finding the utility-maximizing affordable bundle of goods, we find a consumer's demand function for each good (depends on prices and wealth)
- if we want to focus on one good, specifically the relationship between its demand and price, hold all other prices and wealth fixed
- this gives us *individual demand* $q(P)$

Market Demand

- now consider a market with more than just one consumer
- to get a market demand curve for, say, N people, we want to find the total quantity $Q^D(p)$ demanded by the market, as a function of price p
- to get this, simply add up all the individual demand curves:

$$Q^D(p) = q_1(p) + q_2(p) + \dots + q_N(p)$$

where $q_i(p)$ is individual i 's demand

Inverse Market Demand

- sometimes we want the relationship “the other way around”
 - instead of wondering how many units $Q^D(P)$ will be sold at a particular price p ,
 - a firm might want to know the price $P^D(Q)$ at which it *can* sell Q units
- this function $P^D(Q)$ is called the *inverse market demand curve*
- and eg $P^D(5)$ is the “demand price for 5 units”, meaning the price per unit at which the market wants 5 units

Example

Consider a market with 2 individuals. Their individual inverse demand curves are as follows:

$$\text{Individual 1} : P(q) = 10 - q$$

$$\text{Individual 2} : P(q) = 20 - 5q$$

Calculate each individual's demand curve, and also the market demand curve, assuming quantities can't be negative.

Solution for Person 1

- first, invert $P = 10 - q$ as $q = 10 - P$
- the only wrinkle is that this is negative once $P > 10$, and negative quantities aren't allowed
- so let's specify zero demand once $P > 10$
- this gives person 1's demand curve:

$$\begin{aligned} q^1(P) &= \begin{cases} 10 - P & \text{if } P \leq 10 \\ 0 & \text{if } P > 10 \end{cases} \\ &= \max\{10 - P, 0\} \end{aligned}$$

Solution for Person 2

- similarly, invert $P = 20 - 5q$ as $q = 4 - \frac{P}{5}$
- and specify zero demand once this is negative, i.e. $4 < \frac{P}{5} \Leftrightarrow P > 20$
- so person 2's demand curve is:

$$q^2(P) = \begin{cases} 4 - \frac{P}{5} & \text{if } P \leq 20 \\ 0 & \text{if } P > 20 \end{cases}$$

Market Demand

- here we want $Q^D(P) = q^1(P) + q^2(P)$: market demand is the sum of individual demands
- we see that there are 3 ranges to consider:
 - if $P > 20 > 10$, neither person wants anything, so $Q^D(P) = 0$
 - if $20 \geq P > 10$, person 1 wants nothing, so $Q^D(P) = 4 - \frac{P}{5}$ (2's demand)
 - if $P < 10$, both people want to buy, so $Q^D(P) = 4 - \frac{P}{5} + 10 - P = 14 - \frac{6}{5}P$
- SO

$$Q^D(P) = \begin{cases} 14 - \frac{6}{5}P & \text{if } P \leq 10 \\ 4 - \frac{1}{5}P & \text{if } 10 < P \leq 20 \\ 0 & \text{if } P > 20 \end{cases}$$

Equilibrium

- after the midterm, we'll learn how to derive the market supply curve
 - demand reflects the **utility-maximizing** relationship between quantity and price,
 - supply reflects the **profit-maximizing** relationship between quantity and price
- but for the rest of today, we'll simply take the supply curve as given, and explore the basics of equilibrium:
 - what is it?
 - how is it affected by taxes?
 - welfare measures: CS and PS

Equilibrium

- in a nutshell, $\text{supply} = \text{demand}$!
- we'll break this down a little more coming up, but it is always an equilibrium requirement

Example

- consider the demand curve derived a few slides ago

$$Q^D(P) = \begin{cases} 14 - \frac{6}{5}P & \text{if } P \leq 10 \\ 4 - \frac{1}{5}P & \text{if } 10 < P \leq 20 \\ 0 & \text{if } P > 20 \end{cases}$$

- what is the equilibrium price and quantity, assuming supply curve $Q^S(P) = \frac{1}{10}P$?

Example, Solved

- first let's check for an equilibrium with $P \leq 10$
 - this would need $14 - \frac{6}{5}P = \frac{1}{10}P \Rightarrow P = 10.769$
 - this is NOT in the assumed range $P \leq 10$, so no equilibrium here
- next let's check for an equilibrium with $P \in [10, 20]$
 - this would need $4 - \frac{1}{5}P = \frac{1}{10}P \Rightarrow P = \frac{40}{3} = 13.333$
 - this IS in the assumed range $[10, 20]$, **so here's one equilibrium:**
 $P = \frac{40}{3} \Rightarrow Q^D = Q^S = \frac{4}{3}$
- obviously there is no equilibrium with $P > 20$ (supply then exceeds $\frac{20}{10}$ while demand is zero)

Taxes

- next we'll examine equilibrium impacts of taxes
- in this course, we'll just look at simple “quantity taxes”, where the government wishes to collect a flat tax $\$t$ per unit sold

Taxes Paid by Suppliers

- if a quantity tax t is paid (directly) by suppliers, then it shifts their supply curve up by t :
 - supply price $P^S(q)$ is willing-to-sell-for price for the supplier of the q th unit
 - this hasn't changed!
 - so if he has to give t to the government, must collect an extra t from consumers
- thus, new inverse market supply curve is $P^{S'}(q) = P^S(q) + t$

Taxes Paid by Consumers

- if a quantity tax t is paid (directly) by consumers, then it shifts their demand curve down by t :
 - original inverse demand $P^D(q)$ tells you the per-unit price the market is willing to pay for q units
 - this hasn't changed!
 - so if the consumer of the q th unit has to now give $\$t$ to the government, he's willing to give $\$t$ less to the store
- thus, new inverse market demand curve is $P^{D'}(q) = P^D(q) - t$

Equilibrium

- if asked to find the new equilibrium price and quantity, it is most useful to just draw the graphs, and see what happens to the intersection point
- so just remember: tax on suppliers means the supply curve shifts up, tax on consumers means the demand curve shifts down
- **tax incidence theorem:**
 - the equilibrium quantity, and effective (after-tax) price/unit, is the same *regardless of whether consumers or producers pay the tax*
 - but who **suffers** the most depends on elasticity

Example 1

- suppose (inverse) demand is $P^D(Q) = 10 - Q$, while (inverse) supply is $P^S(Q) = Q$
- original equilibrium is $Q^0 = 5, P^0 = 5$
- if **consumers** must pay a \$1/unit tax to the government, then:
 - new demand: $P^{D'} = P^D - t = 9 - Q$
 - this intersects supply where $9 - Q = Q \Rightarrow Q^* = 4.5, P^* = 4.5$
 - so after taxes, suppliers get $P^* = \$4.50$ and consumers pay \$5.50 ($P^* + t$)

Example 1 Cont'd

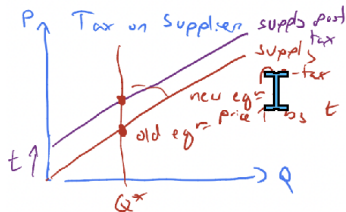
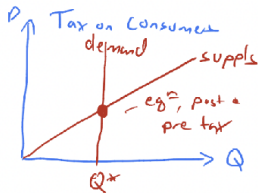
- if **suppliers** must pay a \$1/unit tax to the government, then:
 - new supply: $P^{S'} = P^S + 1 = Q + 1$
 - this intersects demand where $10 - Q = Q + 1 \Rightarrow Q^* = 4.5, P^* = 5.5$
- so after taxes, consumers pay $P^* = \$5.50$ and suppliers take in $P^* - t = \$4.50$
- same after-tax picture as on previous slide!
- tax burden is shared 50-50 here: consumer final price rises 50 cents, and supplier final price falls 50 cents

Example 2: Perfectly Inelastic Demand

- this means that demand is vertical: NO change in quantity demanded if the price changes
 - think of a life-saving drug, where the same quantity is demanded regardless of its price
- again, final quantity traded and effective price is the same regardless of whether it is levied directly on consumers or on suppliers
- but now consumers bear the entire tax burden (see next slide; supply curves are supposed to be parallel!)
 - with a tax on consumers, no change in demand (can't shift a vertical line up!), so no change in eqm price: but now they pay $\$t$ on top of this
 - with a tax on suppliers, supply shifts up by $t \Rightarrow$ it intersects vertical demand at a price that has increased by t

Space for Pictures

Figure for Example 2



Example 3: Perfectly Elastic Demand

- this means that demand is a horizontal line: as an example, say that demand is a horizontal line at $P = 5$ (meaning that consumers are willing to buy any quantity at $P = 5$)
- in this case suppliers bear the full tax burden
- the suppliers will simply get a lower net price per unit, by the amount of the tax